MIL-P-20449C (MU)
18 October 1968
SUPERSEDING
MIL-P-20449B (MU)
15 September 1963 and
MIL-P-46408A (MU)
6 September 1963

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M61
PARTS AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts and the loading, assembling and packing for one type of initiating device designated as Primer, Percussion, M61.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1390

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DRAWINGS

U. S. ARMY MUNITIONS COMMAND

8831157 - Primer, Percussion, M61 Assembly
8861059 - Box, Set-up, Packing, Ammunition, Inner and Outer for Percussion Primer, M61
8861060 - Box, Packing, Ammunition for Percussion Primer, M61

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET - 8831157 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Parts.- The parts shall be in accordance with applicable drawings and specifications.

3.2 Primer assembly.- The primer assembly shall comply with all requirements specified on Drawing (dwg.) 8831157 and with all requirements specified in applicable specifications.

3.3 Functioning.- The primer assembly shall not show evidence of a firing failure (see 6.5) when tested as follows:

3.3.1 Sensitivity.- The primer assembly shall fire without any evidence of failure when a maximum (max.) of one hundred eleven and forty-four hundredths (111.44) inch ounces (see 6.7) of energy is applied and shall not fire when a minimum (min.) of forty-seven and fifty-two hundredths (47.52) inch ounces of energy is applied. The test shall be performed as specified in 4.4.1.1.

3.3.2 Failure to fire (see 6.5).- The primer assembly shall not fail to fire when tested as specified in 4.4.1.2.

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3.4 Workmanship.- All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, cracks, chips, sharp edges, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.5 First article inspection.- This specification makes provision for first article inspection. Requirements for the submission of first article samples shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.- At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.

b. Number of units of product inspected.

c. Results obtained, by defect code, for all inspections performed.

d. Drawing, specification number and date, together with an identification and date of changes.

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e. Statement of findings (certificates of analysis) on the charge and certificates of conformance on all remaining materials purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.- The contractor shall submit a first article sample (see 6.1c) as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of twenty-five (25) each of every component part which has been produced by the contractor or furnished by a supplier and which has been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and he shall submit a record of this inspection with the sample, including statements of findings and certificates of conformance of materials (see 4.1.1e). A sample containing known defects shall not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

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4.2.2 Inspections to be performed.- Components will be subjected by the Government to any or all of the examinations specified in 4.3.2 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.- If any component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.- The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot of primers shall contain:

a. Parts of one lot interfix number from one manufacturer.

b. Primer mixture charge from one batch or blending of batches which results in homogeneity of all primer mixture used in loading the entire primer lot.

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4.3.2 Examination.- Inspection for Critical defects (and Major defects, when so specified) shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cup, primer (see dwg. 8831158 covering a detail of dwg. 8831157).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Outside diameter, min.....	Gage	01001
102.	Inside diameter, max.....	Gage	01002
103.	Thickness through bottom.....	Gage	01003
104.	Total length.....	Gage	01004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4).....	Visual	01005

4.3.2.2 Anvil (see dwg. 8831159 covering a detail of dwg. 8831157).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.65 percent		
101.	Profile	Gage	02001
102.	Diameter, min	Gage	02002
103.	Thickness, min	Gage	02003
104.	Split	Visual	02004
Minor:	AQL 1.00 percent		
201.	Length between grooves	Gage	02005
202.	Evidence of poor workmanship (see 3.4)	Visual	02006

4.3.2.3 Primer, prior to assembling cover (see dwg. 8831157).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Complete charge, or portion of charge missing	Visual	03001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	03002

4.3.2.4 Primer, prior to assembly of anvil (see dwg. 8831157).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	More than one cover	Visual	04001
Minor:	AQL 1.00 percent		
201.	Cover missing, torn, cut or improperly inserted	Visual	04002
202.	Evidence of poor workmanship (see 3.4)	Visual	04003

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4.3.2.5 Primer assembly (see dwg. 8831157).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Cup split or cracked.....	Visual	05001
Major:	AQL 1.00 percent		
101.	Total length, max.....	Gage	05002
102.	Outside diameter, max.....	Gage	05003
103.	Anvil missing, inverted, cocked, loose or damaged to extent that function is impaired.....	Visual	05004
104.	Explosive on exterior.....	Visual	05005
105.	Cup dented or distorted.....	Visual	05006
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship other than specified in Critical above (see 3.4).....	Visual	05007

4.3.2.6 Inner set-up box, prior to sealing (see dwg. 8861059).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Tray damaged to extent that items touch each other.....	Visual	06001
Minor:	AQL 0.65 percent		
201.	Primer missing.....	Visual	06002

4.3.2.7 Inner set-up box (see dwg. 8861059).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			

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Major: None defined

Minor:		AQL 1.50 percent	
201.	Box damaged to extent that contents are exposed.....	Visual	07001
202.	Label missing, mislocated or incompletely sealed.....	Visual	07002
203.	Marking incorrect or illegible..	Visual	07003

4.3.2.8 Outer set-up box (see dwg. 8861059).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent.		
101.	Sealing strip missing, mislocated or incompletely sealed.....	Visual	08001
Minor:	AQL 1.50 percent		
201.	Box damaged to extent that contents are exposed or liable to become exposed.....	Visual	08002
202.	Label missing, mislocated or incompletely sealed.....	Visual	08003
203.	Marking incorrect or illegible..	Visual	08004

4.3.2.9 Outer set-up box sealed in barrier bag (see dwg. 8861059).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Bag cut or torn.....	Visual	09001
102.	Seal incomplete or mislocated...	Visual	09002

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Minor:	AQL 0.65 percent		
201.	Marking missing, incorrect or illegible	Visual	09003

4.3.2.10 Packing box, prior to closing (see dwg. 8861060).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Carton missing	Visual	10001
Minor:	AQL 1.00 percent		
201.	Carton inverted	Visual	10002
202.	Filler missing	Visual	10003

4.3.2.11 Sealed packing box (see dwg. 8861060).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Box damaged to extent that contents are exposed	Visual	11001
102.	Strapping missing, broken or loose	Visual	11002
Minor:	AQL 1.50 percent		
201.	Marking missing or incorrect	Visual	11003
202.	Strapping improperly assembled	Visual	11004
203.	Contents loose	Visual	11005
204.	Evidence of poor workmanship (see 3.4)	Visual	11006

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4.3.3 Testing

4.3.3.1 Functioning

4.3.3.1.1 Sensitivity (see 3.3.1) - Major defect - Code No. 05008.- A sample of four hundred and fifty (450) primers shall be selected from each lot for test. The test shall be performed as specified in 4.4.1.1 using equipment as specified in 4.3.4. The lot shall be rejected if "H average" plus 2.75 "S" multiplied by the maximum weight of the ball exceeds the maximum inch ounce energy value specified, and if "H average" minus 2.25 "S" multiplied by the minimum weight of the ball is less than the minimum energy value specified (see 6.6).

4.3.3.1.2 Duds (see 3.3.2) - Major defect - Code No. 05009.- A sample of three hundred (300) primers shall be selected from each lot for this test. If two (2) or more primers fail to comply with the requirement, the lot shall be rejected. The test shall be performed as specified in 4.4.1.2 using equipment as specified in 4.3.4.

4.3.3.2 Determination of composition (see dwg. 8831157) - Major defect - Code No. 12001.- A sample of not less than one (1) gram shall be selected from each batch or blend of batches for test as specified in 4.4.2. All the composition represented by the sample shall be rejected if the sample fails to comply with the requirements.

4.3.4 Inspection equipment.- Equipment Tabulation Number ET - 8831157 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.3.4.2 Supply and maintenance.- Supply and maintenance of the equipment listed in the ET shall be in accordance with MIL-I-45607.

4.4 Test methods and procedures

4.4.1 Functioning

4.4.1.1 Sensitivity.- The primer shall be assembled in the applicable test fixture specified in 4.3.4. A steel ball weighing 7.94 plus or minus .02 ounces shall be dropped from a height of ten (10) inches on the firing pin. (Drop height shall be measured as the distance from the bottom of the ball to the top of the firing pin.) The number of primers that fire, or fail to fire, shall be recorded. This procedure shall be repeated at successively increasing heights of one inch, using a new group of fifty (50) primers at each height until a height is reached at which all primers in the group being tested fire. The test fixture shall then be set at a height of nine (9) inches and the above procedure repeated at successively decreasing heights of one inch using a new group of fifty (50) primers at each height, until a height is reached at which all primers in the group being tested fail to fire. Acceptance or rejection of the lot shall be determined from the calculations based on recorded data (see 6.6). All samples subjected to test shall be scrapped. Untested samples may be returned to the lot.

4.4.1.2 Duds.- The primer shall be assembled in the applicable test fixture specified in 4.3.4. The steel ball weighing 7.94 plus or minus .02 ounces shall be dropped on the firing pin from a height of not more then fourteen (14) inches.

4.4.2 Determination of composition of primer mixture

4.4.2.1 Trinitrotoluene (TNT).- Transfer approximately 1 gram of the sample to a tared sintered glass or filtering crucible having a thick asbestos mat. Using a wooden spatula, quickly spread the sample in a thin layer over the bottom of the crucible, and weigh rapidly. Allow the crucible to remain at least 16 hours in a desiccator containing calcium chloride, weigh. To the dried sample in the crucible, add 15 milliliters (ml) of anhydrous chloroform. Allow the solvent to stand in contact with the sample for 5 minutes,

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then, with the flat end of a wooden rod, gently and carefully crush the lumps against the asbestos mat. Allow the solvent to drain through the mat without suction. Repeat this extraction with 2 more 15 ml. portions of chloroform. Completeness of extraction may be tested for at this point by adding 1 or 2 drops of the chloroform filtrate, 2 drops of alcoholic potassium hydroxide solution (1 Molar) on a white spot plate. A red coloration indicates incomplete extraction. If extraction is not complete, continue with 15 ml. portions of chloroform until it is as shown by test. Draw air through the crucible until no odor of chloroform remains. Dry the crucible and residue at 100 degrees Centigrade (C). for 15 minutes, cool in a desiccator, and weigh. Calculate the loss in weight to percentage of trinitrotoluene in the sample.

4.4.2.2 Antimony sulfide.- Prepare 25 percent saturated ammonium acetate solution by diluting saturated ammonium acetate solution with 3 times its volume of water. Treat the residue in the crucible from the trinitrotoluene determination with 20 ml. of warm 25 percent saturated ammonium acetate solution. Allow the solution to remain in contact with the residue for 10 minutes, breaking up any lumps with the flattened end of a glass rod. By means of suction draw off the solvent into a clean flask, and repeat the extraction with 2 more 20 ml. portions of warm solution. At the end of the third treatment completeness of extraction may be tested for by treating 1 or 2 drops of the ammonium acetate filtrate with 2 drops of dilute nitric acid and 2 drops of slightly acidified ferric alum solution on a white spot plate. If extraction is not complete, as shown by a red coloration, continue treatment of the residue with 10 ml. portions of the solvent, when extraction is complete, as shown by absence of red coloration when two drops of the filtrate are tested, wash the crucible and residue thoroughly with water, collecting the washings with the ammonium acetate filtrate. Wash the residue with alcohol and ether, and aspirate until the odor of ether is no longer detectable. Then place the crucible with contents in an oven at 100 degrees Centigrade for 15 minutes, and cool in a desiccator, and weigh. Subtract from this weight, the weight of the empty crucible and calculate the difference to percentage of antimony sulfide in the sample.

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4.4.2.3 Lead thiocyanate.- Transfer the filtrate and washings from the antimony sulfide determination to a 400 ml. beaker. The volume should be between 150 and 200 ml. Add 2 ml. of glacial acetic acid and heat the solution to boiling. While stirring vigorously, slowly add 8 ml. of 5 percent potassium dichromate solution. Stir for 1 or 2 minutes longer, then digest on a steam bath for 1 hour, with frequent stirring. The precipitate should have a reddish-orange color. Catch the precipitate on a tared filtering crucible, wash with hot water, and dry the crucible and precipitate for 2 hours at 100 degrees (C.), cool in a desiccator, and weigh. Calculate the increase in weight to percentage of lead thiocyanate in the sample.

Lead thiocyanate = 100.05 x lead chromate

4.4.2.4 Potassium perchlorate plus gum.- Subtract the sum of the percentages of trinitrotoluene, antimony sulfide, and lead thiocyanate from 100 and report remainder as percentage of potassium chlorate plus gum.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.- The percussion primer assembly shall be preserved and packaged in accordance with dwg. 8861059.

5.2 Packing

5.2.1 Level C.- The outer set-up boxes shall be packed as specified on dwg. 8861060.

5.3 Marking.- The containers shall be marked in accordance with dwgs. 8861059 and 8861060.

6. NOTES

6.1 Ordering data.- Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.- Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

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c. Submission of first article quantity.- Instructions as to the location of the facility at which the evaluation of the first article quantity shall be carried out may be obtained from the contracting officer.

6.2 Inspection code numbers.- The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.- The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).- Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.- The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.- Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection

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equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.- All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-NDl. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is

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permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of ammunition data cards.- In addition to the normal distribution of records, one copy of all ammunition data cards shall be forwarded to each of the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey 07801
ATTN: (1) SMUPA-ND5
(2) SMUPA-DC.

6.5 Firing failure.- A firing failure is defined as follows:

a. Misfire or dud.- A failure wherein initiation of the primer mixture does not occur.

b. Squib.- A failure wherein the primer mixture burns with a hiss or fizz.

c. Hangfire.- A failure wherein an audible delay occurs between the application of energy and the detonation report.

6.6 Formula and method of determining H average and S

6.6.1 Formula:

H Average = A plus B plus C
S = I times the square root of L minus M

where:

A = the sum of column 3.
B = the first drop height at which all primers misfired.
C = one-half the drop height increment (see 6.7).
L = the sum of column 5.
M = the square of the sum of column 3.
I = the drop height increment.

6.6.2 Tabulation of data.- Divide a sheet of paper into 5 columns and tabulate the data as follows:

a. Title column 1: "Height of Drop" and list all drop heights in order starting with the lowest height. DO NOT list the heights at which all primers fired or misfired.

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b. Title column 2: "Number of Failures" and list the sum of the number of primers that misfired, squibbed or hangfired (see 6.5) at each intermediate height.

c. Title column 3: "Decimal Fraction of Primer Failing to Fire at Each Intermediate Height" and record the decimal fraction that failed to fire at each intermediate height. This number is obtained by dividing the value obtained under "Number of Failures" by the number of primers tested at that height. Calculation should be carried out and recorded to the nearest one-hundredth (.01).

d. Title column 4: "Standard Deviation Factor" and list odd numerals in sequence starting with 1, for as many drop heights as are listed in Column 1.

e. Title column 5: "Standard Deviation Factor Times Fraction of Failures" and by multiplying each decimal fraction in Column 3 by the number on the same line on Column 4, enter the product on the corresponding line on Column 5.

6.6.3 Procedure for solving equations:

a. To obtain "A" - add the values in Column 3, and substitute the sum for "A".

b. To obtain "B" - the first drop height at which all primers failed to fire and substitute the product for "B".

c. To obtain "C" - divide the drop height increment by 2 and substitute the dividend for "C".

d. To obtain "L" - add the values in Column 5 and substitute the sum for "L".

e. To obtain "M" - square the sum of Column 3 and substitute the sum for "M".

f. To obtain "H average" - add the values substituted for "A", "B" and "C".

g. To obtain "S" - subtract the value substituted for "M" from the value substituted for "L" and then obtain the square root of the difference and then multiply the square root by the height of increment.

h. Using the calculated values for "H average" and "S", if "H average" plus 2.75 "S" multiplied by the maximum weight of the ball is equal to or less than the maximum inch ounce energy value required, and "H average" minus 2.25 "S" multiplied by the minimum weight of the ball is equal to or greater than the minimum energy value required, the lot should be acceptable for sensitivity.

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6.7 Inch ounces.- The product of drop height in inches multiplied by the weight of the ball in ounces.

6.8 Height of increment.- The height of increment is defined as a constant interval in inches between successive drop heights which must be considered in calculating average height and the standard deviation in drop height.

6.9 Check test for deterioration of the M61 Primer.- The following tests are designed to prevent loading M61 Primers which have deteriorated into a major ammunition item. The tests should be performed on primers from each lot which have been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the primers are loaded does not require that the check tests be performed, the contracting officer should insert provisions for the performance of the check tests in the contract. The contracting officer should arrange to have the contractor reimbursed for any expenses incurred in the performance of the check test for possible deterioration.

6.9.1 Sensitivity.- The test shall be performed as specified in 4.4.1.1, using test equipment specified in 4.3.4. A sample of 675 primers shall be selected from each lot for this test. Seventy-five primers shall be tested at each height. If H average plus 2.72 s exceeds 14 inches, or if H average minus 2.22 s is less than 6 inches, the lot shall be rejected.

6.9.2 Duds.- This test shall be performed as specified in 4.4.1.2. A sample of three hundred (300) primers shall be selected from each lot for test. If three (3) or more primers fail to function the lot shall be rejected.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1390-A-749

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PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

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343-221 (A-594)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE	
CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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